

Scales

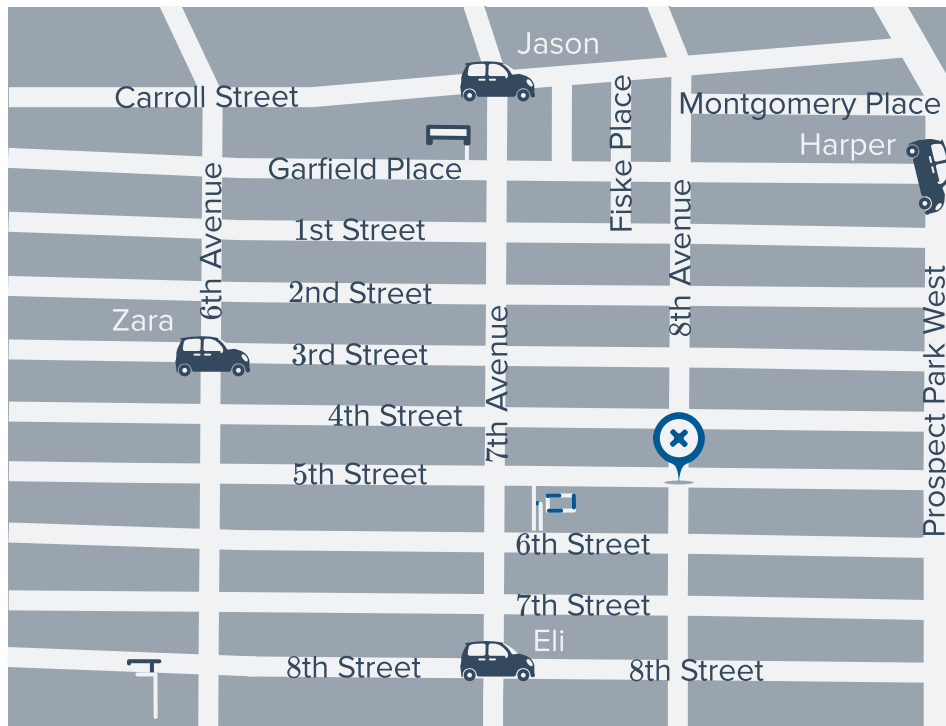


- 1 Convert the following descriptions to a scale ratio:
 - a 3 cm represents an actual distance of 9 m.
 - b 20 cm represents an actual distance of 16 km.
 - c 8 mm represents an actual distance of 40 m.
 - d 60 mm represents an actual distance of 5400 m.
 - e 5 cm represents an actual distance of 30 km.
 - f 98 mm represents an actual distance of 9.8 km.
- 2 The scale on a map of a garden is 1 : 2000. How far apart on the map should two fountains be drawn if the actual distance between the fountains is 100 m? Express your answer in centimetres.
- 3 The scale on a map is 1 : 400 000. How far apart on the map should two train stations be drawn if the actual distance between the stations is 100 km? Express your answer in centimetres.
- 4 Consider a map with a scale of 1 : 2000.
 - a Find the actual distance (in centimetres) between two points which are drawn 19 cm apart on the map.
 - b Find the distance between the two points in metres.
- 5 Consider a map with a scale of 1 : 25 000.
 - a Find the actual distance (in centimetres) between two points which are drawn 14 cm apart on the map.
 - b Find the distance between the two points in kilometres.
- 6 Bianca is looking over a map of her local area and notices that the scale of the map is given as 1 : 100 in the map legend.
 - a Find the actual distance (in centimetres) between two points which are drawn 12 cm apart on that map.
 - b Find the distance between the two points in metres.



Maps and scale diagrams

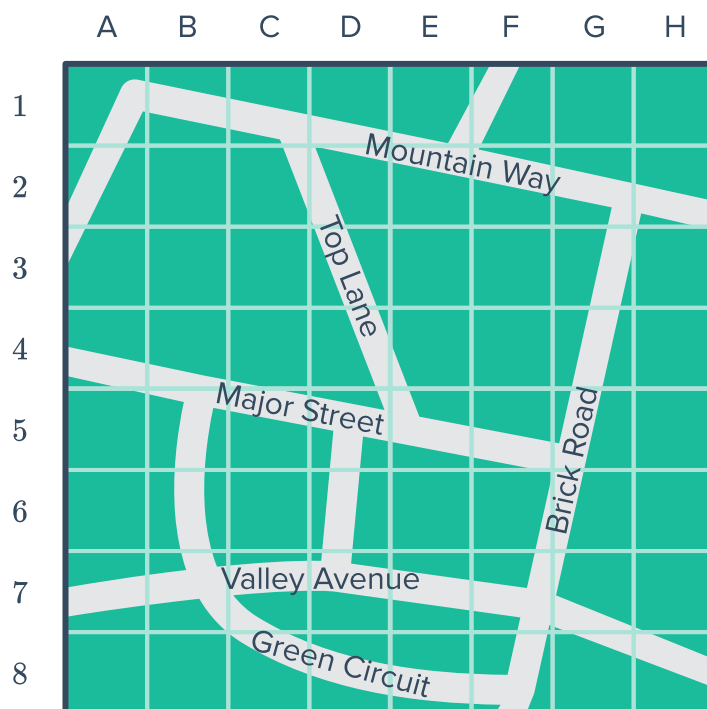
- 7 Below is a map of New York. The actual dimensions of one rectangular block is 230 m by 80 m.



- a How many metres will Zara need to drive to reach the pick-up point, marked with an x?
- b Whose car is the closest to the pick-up location?
- 8 Use the map below to find the distance between Adelaide and Brisbane:



- 9 This is a map of where Adam and Carl live.  side of a small square on the map represents 100 m:



Adam is on the corner of Top Lane and Major Street (map reference *E5*). He walks 300 m along Top Lane to Carl's house.

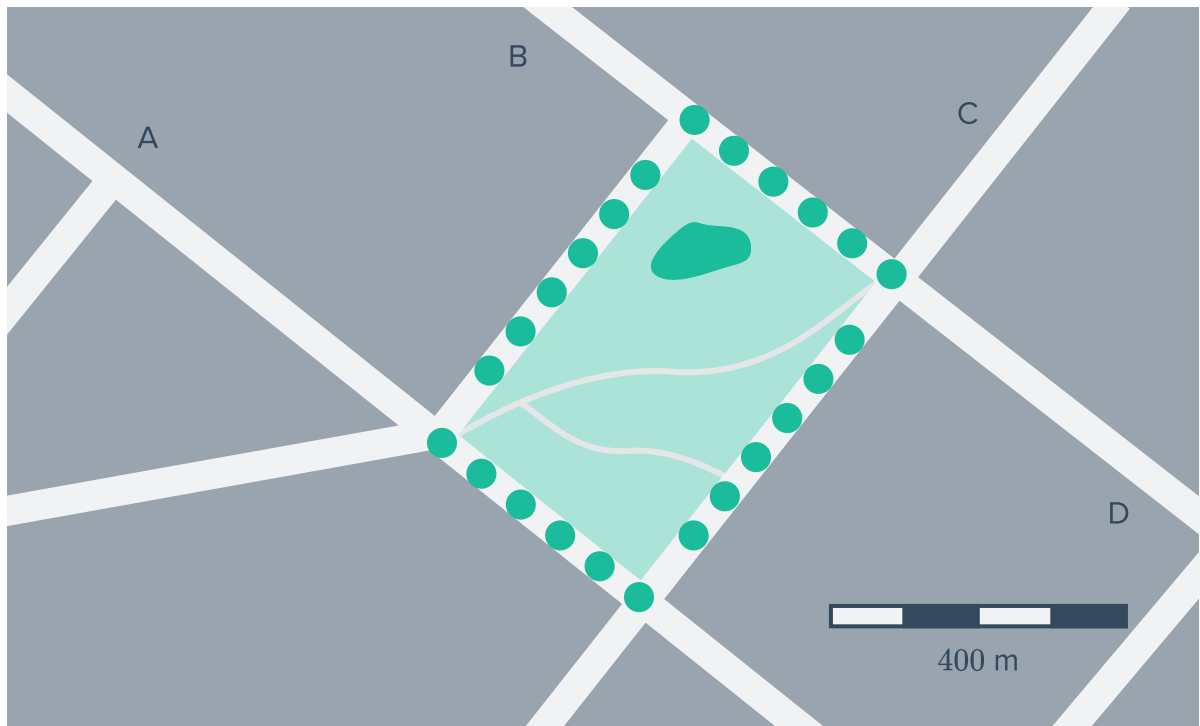
Write the map reference of Carl's house.

- 10 The map designer for a new amusement park measures the main street to be 4 cm. The walk along the main street is actually 120 m:



Write down the scale ratio of the map.

11 A map of a town is drawn to scale below:



- a Find the distance between house A and the park.
- b Find the distance between house C and the park.
- c Find the length of the shorter side of the park.
- d Find the length of the longer side of the park.
- e Find the distance traveled by driving from house A to house C along the roads.
- f Find the distance traveled by first driving from house A to house C, then driving to house B, traveling along the roads.

12 The following is a 1 : 58 000 scale drawing of the sailing route from the mainland to an island off the coast. The captain approximates the distance to be 15.3 cm on the map. Estimate the distance in kilometres of the boat trip.



- 13 Estimate the distance from Perth to Brisbane using the map scale:

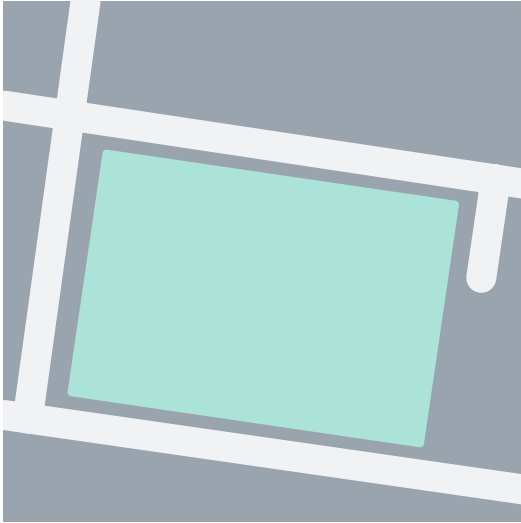


- 14 The scale on the map of a public garden is 1 : 5000. How far apart on the map should two ponds be if the actual distance between the ponds is 350 m? Give your answer in centimetres.



- 15 Vincent is comparing three parks from three different maps shown below. Which park has the largest area? Explain your answer.

Lesse park



Scale: 1 : 2000

Millenial park



Scale: 1 : 5000

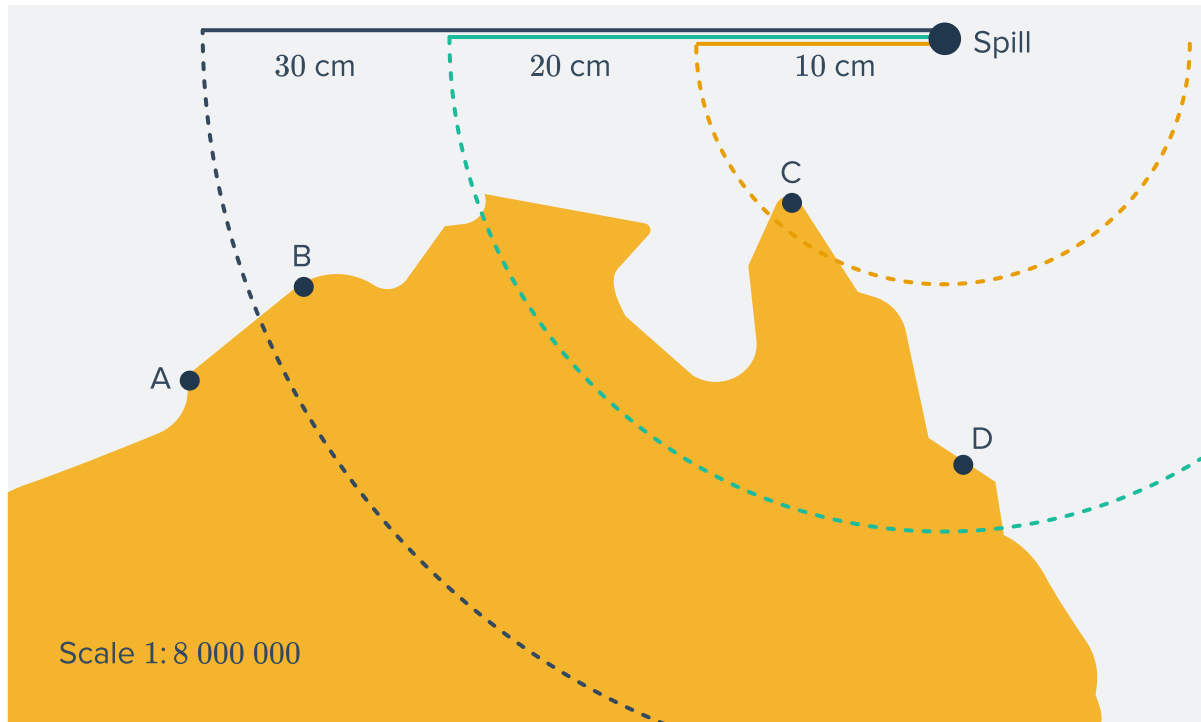
Tibsen park



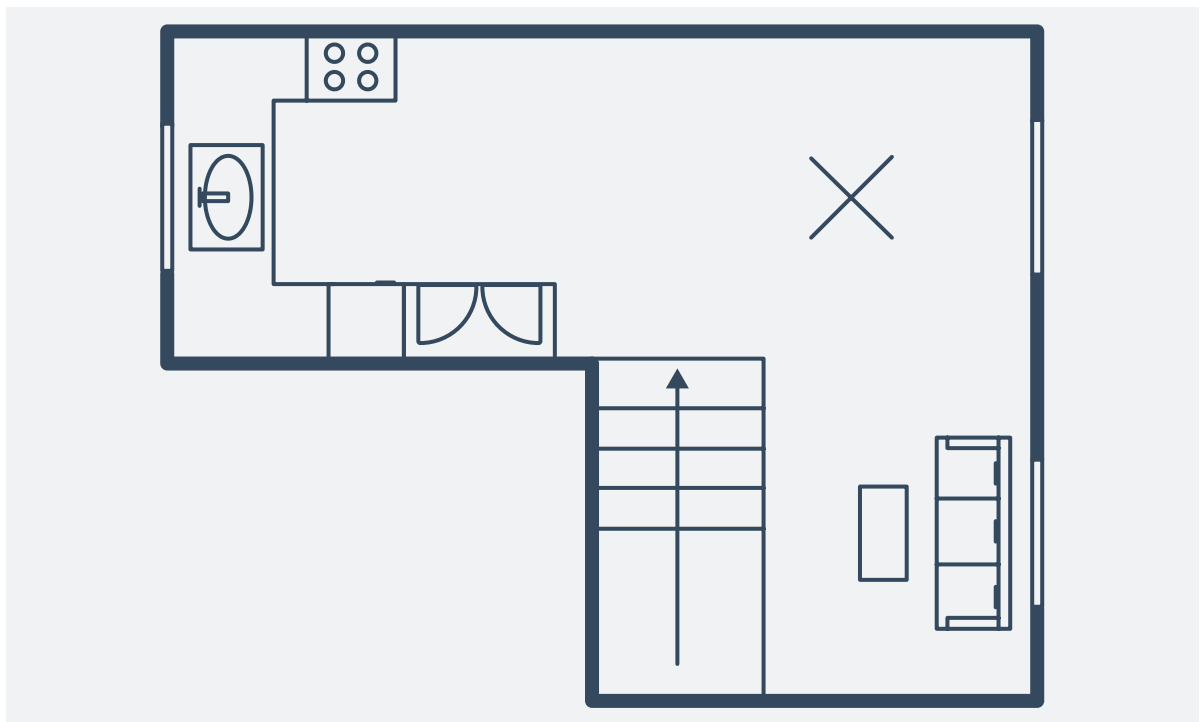
Scale: 1 : 7000

- 16 A map is being drawn for a medium-sized rural town. The map will fit on an A4 piece of paper which has the dimension of 21 cm by 29.7 cm. Write an appropriate scale for the map.

- 17 An oil tanker has a spill off the coast of Australia. An expert says that the oil could travel 1700 km in the first day. Their map is shown sketched with circles with radius 10 cm, 20 cm and 30 cm:

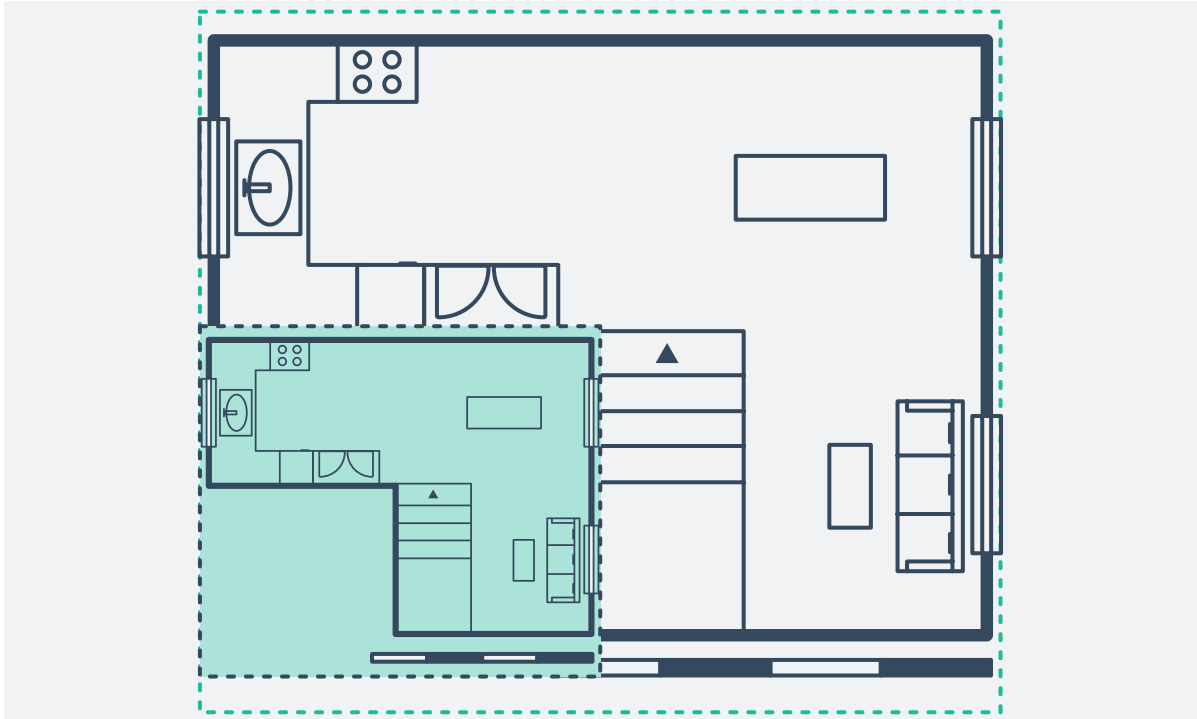


- Which locations will be affected in the first day?
 - Location B is 26 cm from the oil spill on the map. Find the actual distance of this location from the oil spill.
- 18 The following is a 1 : 200 floor plan of a house. The homeowner wishes to add a dining room table, which is 150 cm long, placing it where the × is marked on the floor plan.





- 19** A plan of a building with ratio 1 : 90 is enlarged so its length and width are both doubled as shown:



- a** Find the new scale ratio for the larger plan.
- b** The scale bar on the original plan represents 5 m. When the plan is doubled, the scale bar is doubled as well. Find the length that the new scale bar represents.
- 20** An architect is designing a house that is 21 m long and 15 m wide.
- a** She wants to print the plans for this house in a 84 cm by 60 cm section of a large piece of paper. Find the ratio she should use to fit the plans on the paper.
- b** After printing the plans, the architect measures the length of the hallway to be 24 cm on paper. She wants to order a rug that covers the full length of the hallway.
- How long should the rug be in metres?